

Yasith Silva

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OBJECTIVE	I am a dedicated electronic engineering student, passionate in Embedded Systems Design, Circuits Design, IoT Systems and Digital Systems Design . As a team player with good leadership qualities, communication and presentation skills, and experience, I can adapt and learn quickly. I seek opportunities to contribute to the ever-evolving and exciting world of embedded systems, learning along the way.	
TECHNICAL SKILLS	Skills: Circuit Design - Schematics and PCBs, Embedded Systems Design, Programming, Firmware Development (With Mistra Standards), Digital Systems Design, FPGA Implementation Circuit Design: Altium Designer, KiCAD, High speed design, Multilayer Design, Hierarchical Schematic Design Embedded Systems: Embedded C, Baremetal, FreeRTOS Programming Languages: C/C++, Python, JavaScript Other: GitHub, Experience using Linux (Ubuntu, Raspberry Pi OS)	
EXPERIENCE	Trainee Associate Electronic Engineer	<i>Zone24x7 (pvt) ltd</i> Dec 2024 - Jul 2025
	- Worked on IoT, UWB localisation and Embedded System - Coded firmware for multiple projects following industry standards - Got hands on experience with complex embedded systems - Developed presentation skills through student visits, client demonstrations	
	Course Instructor - Embedded Product Design for IoT	<i>Skillsurf</i> Dec 2024 - May 2025
	- Did sessions on schematic and PCB design	
	Freelance PCB Designer	May 2024 - Present
	- Designed and Developed Custom PCB Layouts for projects using <i>Altium Designer</i> - Communicated effectively with clients to understand project requirements	
	Visiting Instructor	<i>University of Moratuwa</i> Dec 2024 - May 2025
	- Assisting telecommunication laboratory practices - Managing the instructors for telecom practicals	
	Co-founder	<i>Metronix</i> Jan 2025 - Present
	- Looking into UI/UX design and marketing	
EDUCATION	University of Moratuwa, Sri Lanka <i>BSc. (Hons) Engineering, Electronic and Telecommunication</i> May 2022 - Present	
		CGPA: 3.67/4.00

D.S. Senanayake College, Colombo 07, Sri Lanka
G.C.E. Advanced Level (Physics, Mathematics, Chemistry)
2012 - 2020

3As / z-score 2.21

Maris Stella College, Bolawalana - Negombo, Sri Lanka
2007 - 2012

COURSES AND CERTIFICATES

Introduction to Embedded Machine Learning — Edge Impulse
FPGA Softcore Processors and IP Acquisition
University of Colorado Boulder
Hardware Description Languages for FPGA Designs
University of Colorado Boulder
Introduction to, Intermediate Machine Learning — Kaggle
MATLAB Onramp — MathWorks

PROJECTS

Chip-Aware Universal Evaluation Platform (Ongoing):

FYP. A universal platform that can be reconfigured to evaluate a range of ICs, including motor controllers, ADCs, DACs, and the software required for configuring, and an AI agent to convert the datasheets to configuration files and assist testing.

- Designing the schematics, PCB and firmware for the expansion kit for the FPGAs, containing an ADC, a DAC, gain controllers, expanders and other inputs and outputs for the device under test.
- Designing the test cases

METROBAND — A Metronome Wristband:

An alternative to the traditional metronome which aids musicians to keep to their tempo, using vibrations. The circuit was designed around an ESP32-S2 chip, communicates with the phone using Bluetooth.

- Designed with *ESP32-S2-WROOM*
- Developed the firmware
- Schematics and PCB Design done using *Altium Designer*

Wireless Reconfigurable Andon System with Maintenance Prediction:

A System to flag problems in manufacturing lines. We equipped this with wireless data logging and maintenance prediction.

- Designed with *ESP32-S3-WROOM* chip
- Baremetal C coding with *ESP-IDF*
- Schematics and PCB Design using *Altium Designer*

EcoWatt - Energy Monitoring Gateway:

A microcontroller-based energy monitoring gateway simulating a solar inverter interface, designed with constrained uplink, remote configuration, security, and FOTA support.

- Wrote firmware for remote configuration
- Managed refactoring the code to abstraction layers and keeping in Mistra standards

Flood-level Sensing System for Detection and Prediction:

A system with several sensors to sense the water level of a body, and a rain gauge to predict the water level, and sends the collected data from a close area to a central server.

- Designed with *ESP12-E* chip
- Uses *MQTT* over Wifi to communicate with the server
- Schematics and PCB Design using *KiCad*

A Task-Oriented Robot:

An autonomous line follower robot, with abilities of collision avoidance, a mechanical arm able to pick up an object, colour detection, sound detection, detecting a moving guard robot, and avoiding it

- Designed with *Arduino MEGA 2560*

32-bit RISC-V Processor:

A 32-bit soft-core processor on an FPGA to support RISC-V. Consists of the basic peripherals required to function.

- A single cycle processor designed first, and will be expanded to be pipelined
- Contains cache and branch prediction.

UART Transceiver using DE0-Nano FPGA Board:

UART Transceiver, communicating between two FPGA Boards. The design was modified to control a series of LEDs on each board from the other.

- Implemented on an *DE0-Nano* FPGA Board
- Design verified by a Simulating a Testbench on *Altera-ModelSim*

Guitar Pedalboard:

A set of fx pedals such as Overdrive, Fuzz, Tremolo and Wah. This was done as a fully analogue project, using operational amplifiers.

- PCB Design using *Altium Designer*
- Assembled and soldered the components

COMPETITIONS VRCade 2.0 (Winners)

Signify — A text-to-sign language application for ease of communication for hearing impaired, designed for an AR glass, and a customized lightweight AR glass able to specifically run the AR application.

- Designed and customized the PCB for the AR Glass

DVCon2025 India (First Runners Up)

A Custom Hardware Accelerator for U-Net — An accelerator for image segmentation using U-Net for Autonomous Driving Applications. This is to be integrated with the VEGA AT1051 - 32bit CPU IP core, running on a Genesys-2 board.

- Made the U-Net IP

Idealize'24 (First Runners Up)

Spark Challenge 2023/24 (Finalists)

SportSense — A mobile app that helps users to keep correct poses and forms while doing their exercises using a body pose landmark identification model.

- Contributed in UI/UX design with *Android Studio*

Brainstorm'24 (Finalists)

Project CrystalClear — A platform that consists of interactive exercises for Dyslexic patients using a finger-pose model evaluation.

- Contributed in UI design using *Codux*

Uva Wellassa University Robot Battles 2.0 Death Race (Semifinalists)

A Remote Controlled Battlebot on several races across an outdoor path containing various obstacles such as ramps, saws, hammers and fire.

- Designed the PCB and the wiring system
- Contributed in coding the RF communication part

SOFT SKILLS

Communication Skills

- Language Skills
 - English — Bilingual proficiency
 - Sinhala — Native proficiency

- Presentation Skills
 - An active member of the Zone24x7 Toastmasters Club
 - Presented company projects at Zone24x7 at multiple student visits and in client demonstrations
 - Was one of the presenters at the Mobitel Lab of the department during the EXMO-2023 exhibition.
 - Presented Metroband—The metronome wristband at EXMO-2023
 - Have been a presenter in several other project presentations and in competitions

Leadership and Teamwork

- In charge of the telecom lab instructors
- Was the leader of several group projects at the university.
- Was a batch coordinator for the Mobitel Lab at the EXMO-24 exhibition
- Was a part of the PR team of SLRC-2024
- Was one of the PR-IT Directors at the Rotaract Club of Achievers Lanka Business School for 2021/22
- Been on the organising committees at the school events.

VOLUTEERING AND CLUBS	PR & IT Director — Rotaract Club of Acheivers Lanka Business School	2021/22
	Video Editor — Electronics Club	2023
	Member of Zone24x7 Toastmasters Club	
	Member of Association for Computer Machinery - University of Moratuwa	
	Member of IEEE Student branch - University of Moratuwa	
	Member of Classical Music Society - University of Moratuwa	

During School

President — Catholic and Christian Society of D.S. Senanayake College	2019
Co-organizer — Aeronautical Society of D.S. Senanayake College	2019
Co-organizer — Western Music Society of D.S. Senanayake College	2019
Member of Senior Boys' Choir of D.S. Senanayake College	

REFERENCES

Dr. Ajith Pasqual *Ph.D. (Tokyo)*

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Dr. Subodha Charles *Ph.D. (Florida)*

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